Synthesis of QDs

In QDs, electrons are confined in all three dimensions, but there are also other quantum-confined semiconductors. Quantum wires confine electrons and holes in two dimensions, while the third dimension is free to propagate. Quantum wells confine electrons or holes in one dimension and allow free propagation in the other two dimensions. Many semiconductor substances can be used as QDs, such as cadmium-selenide, cadmium-sulfide or indium-arsenide.

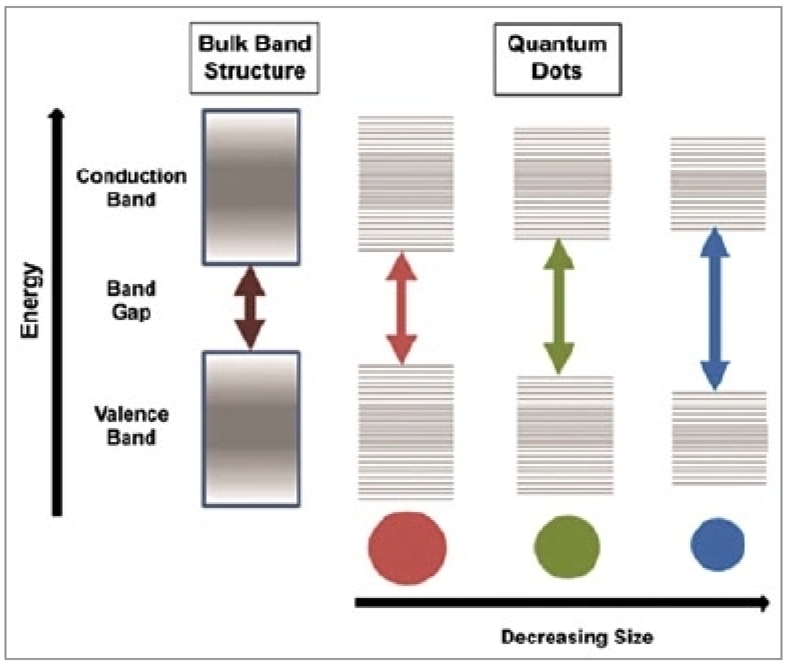

Fig. 2: Semiconductor band gap increases with decrease in size of the nanocrystal (Credit: www.sigmaaldrich.com)

Synthesis of QDs require a combination of an appropriate metallic, or an organo-metallic precursor such as zinc, with a corresponding chalcogen precursor such as sulphur. The solvent must be stable at high temperature to prevent particle aggregation. Also, it must act as a surfactant molecule for the stabilisation of QD surface.

Hossam Haick, head - Laboratory for Nanomaterial-Based Devices, Department of Chemical Engineering at Russell Berrie Nanotechnology Institute, explains, "QDs must be produced in an inert atmosphere due to their reactivity. To protect QDs, selection of ligands depends on the desired application of QDs and their dispersion media."

He adds, "Different methods to stabilise luminescence properties of semiconductor QDs in aqueous media include surface preservation with protective layers, such as proteins and coating of QDs with silicon oxide films, polymer films and so on."

Sensing and imaging with QDs

Upon heating, QDs undergo significant changes in both luminescence intensity and peak position for a short period. And QD device luminescence in density gradually decreases. Corresponding change in colour is also seen with rise in subsequent temperature.

Optical properties of QDs are related to their size and composition. For biological applications, tuning of optical properties is important. Colour of QDs can be used as a diagnostic tool for sensing applications, to integrate QDs and drug molecules into nanoparticle formulation, conjugate or link drug molecules to QD surface.

Applications of QD technology

Dr Manish Kumar Gupta, assistant professor, Symbiosis University of Applied Science, says, "QDs technology can be used for a wide range of applications such as QLEDs, medical imaging, disease diagnostic, solar cells and photovoltaics, defence, memory devices (quantum computing), humidity and pressure sensors."

Some common applications of QDs are discussed below.

- QDs are used to detect the presence of biomolecules. For example, these can detect the presence of sugar maltose by conjugating the maltose-binding protein with the nanocrystal surface.
- QDs can also be used for sensing specific DNA sequences.
- High resistance to metabolic degradation and fluorescence properties enable a wide range of experiments.
- QDs' high photostability and brightness make these suitable for high-sensitivity applications like leaf cell imaging.
- Temperature-dependent changes in optical properties of QDs is the basis for temperature measurement technique.
- Because of their highly-tunable properties, there are various potential applications in transistors, solar cells,